

Developmental and Clinical Correlates of Cerebral Dysrhythmia

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IN THE absence of specific neurophysiological correlates of emotional illness, it has been difficult to delineate syndromes of cerebral dysfunction which might be related to psychiatric disorders. The armamentarium of the investigator in this area is usually limited to clinical- and psychological-test examination of the patient, a variety of neuropsychological tests such as those devised by Halstead and Reitan,^{1,2} critical flicker fusion, the electroencephalogram (EEG), psychophysiological tests such as those devised by Lacey and Lacey,^{3,4} and response to psychotropic or anticonvulsant agents.⁵ Family, developmental, and social histories have received relatively little attention although more recent work has been done in these areas relative to cerebral function.

Although clinical correlates of the EEG have been difficult to elucidate in the past, more recent work has indicated that the EEG may be useful as a predictor of developmental and clinical characteristics of psychiatric patients. The current study aims at delineating developmental, behavioral, affective, and cognitive correlates of EEG dysrhythmias. In particular, the EEG pat-

terns to be studied include those which can be categorized as without dysrhythmia and within normal limits (NEEG), the 14 and 6 cycles per second positive spike pattern (PSP) of Gibbs and Gibbs,⁶ and occipital slow-wave (θ) activity, usually predominant on the right (ROC), first described by Aird and Gastaut.⁷

Both the PSP and ROC have been implicated by a number of investigators as bearing relationship to some aspects of psychiatric illness, usually behavior disorder or other character abnormalities.

Reviews of the PSP have been done by Henry,⁸ Hughes,⁹ Hughes et al.,¹⁰ and Greenberg.¹¹ Although no specific *behavioral* correlates of the PSP have been delineated, there appears to be a significant number of psychiatric patients under the age of 30 who have this EEG pattern, often accompanied by affective remoteness and a lack of empathic responsiveness. The affective remoteness has also been described by L. L. Burnett (unpublished data). Henry⁸ and Greenberg¹² have suggested that the clinical correlates of the PSP may be heavily influenced by sociofamilial factors such as economic deprivation and a disrupted, isolated home environment.

Less investigation has been done on the ROC pattern, although Cohn and Nardini¹³ described dyssocial behavior in patients with the ROC. Greenberg and Pollack¹⁴ described defects in visuomotor performance as mea-

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sured by the Wechsler Adult Intelligence Scale (WAIS), particularly the object-assembly subtest of the performance scale.

Purpose.—The present study aims at testing the following hypotheses: (1) There is a significantly higher incidence of developmental abnormality in patients with EEG dysrhythmia than in those with normal EEGs. (2) There are behavioral differences between patients with cerebral dysrhythmia and those without dysrhythmia. (3) There are affective differences among the three groups of patients, the PSP group in particular being more underemotional than the other two groups (ROC and NEEG). (4) There are cognitive differences, as measured by the WAIS, among the groups, the ROC group having the poorest visuomotor abilities.

Setting.—The setting for this study is a small suburban psychiatric hospital dealing primarily with patients with functional psychiatric disorders who require hospitalization. All patients enter voluntarily and ability to pay is not a factor in determining admission. Almost all modalities of individual and social psychiatric therapies are available.

Method

A consecutive sample of admitted patients between the ages of 16 and 30, over an eight-month period, were withdrawn gradually from psychoactive medication within five to seven days after admission. During the patient's first month in the hospital the resident psychiatrist obtained a full history and became actively engaged in studying his patient. During the fourth or fifth week of hospitalization, prior to any administration of psychoactive medication, the following data were collected for each patient.

Electroencephalogram With the Patient Awake and Asleep.—This included a special montage devised by Henry⁸ which records the 14 and 6 cps positive spike phenomenon.

A Structured Interview Schedule.—This schedule was followed by all residents and social workers in collecting historical data from the patients and their families for patient's charts. These charts were reviewed by two independent chart reviewers, one without knowledge of the EEG data, for the following items: complications of pregnancy, complications of the neonatal period, speech and motor

abnormalities in childhood, neurotic or aggressive behavior or both in childhood, serious childhood illness, and shy socialization in childhood. The ratings were dichotomous and there was better than 95% agreements in all items.

Special Behavioral Questionnaire.—This included items specifically constructed to rate the behavioral and affective variables germane to this study. The questionnaire was filled out for each patient by his resident, who had no knowledge of EEG or WAIS results at the time of data collection. A resident assigned to the study, who was unaware of other findings, helped the other residents fill out the questionnaire. An independent preliminary reliability study showed all items on the questionnaire to have acceptable inter-rater reliability coefficients ($r = 0.85$ or higher).

Full-Scale WAIS.—This test was administered by the clinical psychology staff of all patients.

Family Interaction Questionnaire.—This was filled out by the social worker for each patient's nuclear family of origin. Bias was absent here, since the worker knew nothing of the EEG findings.

Electroencephalographic Data.—Of the 125 patients between the ages of 16 and 30 admitted to the hospital, 121 patients had an EEG taken; 107 of these had a sleep tracing as well as a waking record. The EEGs were recorded on an EEG machine (Grass Model III), using needle electrodes in standard bipolar montages. Methypylon (Noludar), in amounts of 200 to 500 mg, was frequently used to induce sleep, since it does not induce fast activity nor obscure the positive spike pattern. The sleep records were obtained in a special montage devised by Henry⁸ in which occipital and temporal electrodes are each recorded with the contralateral ear as the reference electrode.

Of the 107 appropriate cases with records taken both awake and asleep, 46 records were normal, 28 records showed the 14 and 6 cps positive spike phenomenon as the only abnormality, and 8 showed right posterior slow-wave activity as the only dysrhythmic findings. The other 25 records revealed non-ROC slow-wave activity with or without positive spiking. Twenty-two patients with normal records and 26 patients with positive spiking had all other data available. All eight patients with occipital slow-wave activity had most other data available, the only missing data being in the developmental variables for two patients.

Reliability of the EEG readings was done as part of an overall hospital EEG reliability study by two independent readers, the hospital neurologist (reader 1) and an experienced pe-

Table 1.—Means for Demographic and Intelligence Quotient Data*

EEG Type	No. of Males	No. of Females	Age	Yr of Education	Full-scale IQ	Verbal-scale IQ	Performance-scale IQ
Positive spike pattern	9	7	19.8	12.4	111.8	117.2	103.4
Normal EEG	8	6	20.5	12.6	113.5	119.6	103.4
Right occipital slowing	4	4	19.8	12.9	107.5	116.3	94.9

* No significant differences among groups.

diatric neurologist and electroencephalographer at another hospital (reader 2). Of the 13 cases read by reader 1 as showing positive spikes only, all were read as showing positive spiking by reader 2. Of the 11 records read as normal by reader 1, nine (82%) were read as normal by reader 2. All records were also read by a third EEG reader to ascertain inter-reader reliability for ROC. Both readers agreed that all eight records showed occipital slowing.

Family Data.—Because, in a previous study,¹³ it was found that many of the developmental, behavioral, and WAIS items were strongly influenced by pathological sociofamilial backgrounds, only patients who were not from severely disturbed family backgrounds served as subjects in this study. A full description of the method of ascertaining the degree of disturbance in the sociofamilial background has been given by Greenberg.¹¹

A family was considered to have the asocial family syndrome (AFS) if, prior to the patient's puberty, the following signs of sociofamilial dysfunction were present: (1) social isolation; (2) significant psychiatric disease in a parent or medical illness in a member of the patient's nuclear family; and (3) signs of actual disruption of the home or high degree of emotional distance or hatred between parents.

The special family questionnaire filled out by the social workers was used to exclude patients with such families from the current study. Thus, the final sample had 14 patients with normal EEGs (NEEG), 16 patients with the positive spike pattern (PSP), and 8 patients with right posterior slowing (ROC), all from relatively intact sociofamilial backgrounds. Socioeconomic status was the same for all EEG groups in this sample, almost all patients coming from comfortable middle-class backgrounds.

Results

Demographic Data, WAIS IQ Findings, and Diagnoses.—The demographic data and full-scale intelligence quotient (IQ), verbal-scale IQ, and performance-scale IQ are given in Table 1. Although there were no

significant differences in the IQ scores, the ROC group displayed a marked trend toward a lower performance-scale IQ.

Sixty percent of the patients had diagnoses of schizophrenia and the rest were diagnosed as having character disorders. There were no significant differences between groups with respect to diagnosis.

Developmental Findings.—Significant differences between groups were found for items related to complications of pregnancy, neurotic or aggressive behavior or both in childhood, and serious illness in childhood (Table 2); no differences were found in the incidence of neonatal complications, speech and motor development, and social isolation.

For the items in which significant differences were found the specific results are:

There were significantly more complications of pregnancy in the ROC group than in either of the other two groups, although the NEEG and PSP groups did not differ on this variable. Thus, five of 6 ROC patients (87%) had histories of complications of pregnancy, while six of 16 PSP (37%) patients, and two of 12 NEEG patients (17%) had such complications.

The PSP group exhibited the highest incidence of neurotic-aggressive behavior in childhood; there were no differences between the ROC and NEEG groups. Specifically, none of the six ROC patients (0%), 13 of the 16 PSP patients (81%), and seven of the 14 NEEG patients (50%) exhibited neurotic-aggressive behavior in childhood.

The most striking differences between groups appeared in the frequency of significant illness in childhood. No patients in the ROC group (0%) had serious childhood illness and only one of the 14 patients in the NEEG group (7%) had such illness. Conversely, eight of the 16 patients on the PSP group (50%) had serious illness in childhood, including high fever in two cases

Table 2.—Fisher's Exact-Test Probability Values for Developmental Data

Item	ROC-PSP	ROC-NEEG	NEEG-PSP
Incidence of complications of pregnancy	0.05	0.025	NS
Neurotic-aggressive behavior in childhood	0.005	NS	0.06
Incidence of serious illness in childhood	0.01	NS	0.02

with delirium or convulsions, asthma, colitis, and head trauma with resulting unconsciousness. The PSP group had significantly more illness than did the other two groups.

Affective Findings.—The affective variables rated were (1) underemotionality; (2) hyperaffectivity; and (3) blunted affect.

An affective cluster score (C_1), composed of the totals of these three items, was computed for each patient. The affective findings were analyzed by analyses of variance and *t*-tests and by using a dichotomous classification on each item evaluated (χ^2 and Fisher's Exact Tests). Analyses for validity and inter-item correlations are found elsewhere.¹¹

Although none of the *F*-tests were significant on analyses of variance among the three groups, one-tailed *t*-tests revealed differences in the predicted directions such that the PSP patients had significantly more underemotionality and greater overall affective deficit in the affective cluster (C_1) and had more blunted affect than did the ROC patients. The differences between PSP and NEEG groups, while having a trend toward such findings, did not reach significance.

There were no differences between groups in hyperaffectivity. The overall affective cluster (C_1) when dichotomized at a cutoff point described elsewhere,¹¹ showed a χ^2 significant at the 0.05 level. The PSP group had the greatest affective deficit. The same trend, although it failed to reach significance, appeared between the PSP and NEEG groups. When the overall affective cluster (C_1) was broken down into its components of underemotionality, blunted affect, and hyperaffectivity, no significant differences at the 0.05 level could be found.

Behavioral Findings.—No differences be-

tween or among groups could be found for overt aggressive behavior, nor for manipulativeness or dependent behavior although there was a trend for the PSP group to be more overtly aggressive than the ROC group.

Intellectual Function.—As previously stated, there were no differences between groups for the full-scale IQ, verbal-scale IQ, or performance-scale IQ. Moreover, there were no differences between groups in the verbal-scale subtests. As can be seen in Table 3, there were several differences between groups on the performance-scale subtests. There were also significant differences on some special combined scores.

Although there were no significant *F*s for the performance-scale subtests, the ROC group showed significantly lower block-design, picture-arrangement, and object-assembly scores than did the PSP group, using the *t*-tests. Block design and object assembly were both evaluated on a one-tailed test since findings were predicted in one direction; the picture arrangement was evaluated on a two-tailed test since no prediction had been made. In addition, as has been previously found, the ROC group scored significantly lower than did the NEEG group on object assembly.

Special Combined WAIS Subtest Scores.

—Six combined scores were evaluated for differences between groups. These were the verbal, attention, and perceptual-analytic triads of Witkin et al.¹⁶ (Verbal triad = Sum of scale-scores of information, comprehension, and vocabulary. Attention triad = Sum of scale-scores of arithmetic, digit span, and digit symbol. Perceptual-analytic triad = Sum of scale-scores of picture completion, block design, and object assembly) the difference between information and block (I-BD), and the difference between information and object assembly (I-OA). In addition the overall difference between verbal-scale IQ and the performance-scale IQ was analyzed.

As predicted, there were no significant differences between groups found on the verbal or attention triads. Although the *F*-test was not significant for the perceptual-analytic triad, the ROC mean score was lower than those of the NEEG and PSP groups.

When the information scale score was

Table 3.—Means and *t*-tests for Intellectual Data (WAIS)

EEG Type	Verbal Scale Subtests*						Performance Scale Subtests				
	Inf	Comp	Arith.	Simil	Dig Span	Vocab.	Dig Sym	Pict Comp	Blk Des	Pict Arr	Obj. Assem.
Positive spike pattern	12.98	12.56	11.25	13.06	11.09	13.69	11.56	10.00	12.06†	11.19‡	10.38
Normal EEG	14.00	13.21	11.51	13.14	12.29	13.87	10.57	10.14	10.43	11.14§	10.07
Right occipital slowing	14.00	12.75	11.13	12.38	11.13	13.37	11.50	9.13	8.25	9.38	7.13

* No significant differences among groups.

† $t = (1, 3)$: $P < 0.05$ (one-tailed).

‡ $t = (1, 3)$: $P < 0.05$ (two-tailed).

§ $t = (1, 3)$: $P < 0.01$ (one-tailed).

|| $t = (2, 3)$: $P < 0.05$ (one-tailed).

compared with block-design and with object-assembly scale scores, significant results were found. Because information is the most reliable and least spuriously elevated subtest in the verbal triad it was used as a standard of comparison. Thus, the *F*-tests for both information minus block design (I-BD) and information minus object assembly (I-OA) were significant at the 0.05 level. The *t*-test (two tailed) results indicate that for the ROC group, the mean differences between information and block design (I-BD) on the one hand, and between information and object assembly (I-OA) on the other, are greater than for the PSP groups at the 0.01 level of confidence. There were no significant mean differences for I-BD and I-OA between the ROC and NEEG groups but there were strong trends in the direction of greater discrepancies in the ROC group. Analysis of the data using a Mann-Whitney U-test yielded, however, significant differences between NEEG and ROC groups for both variables, I-OA being significant at the 0.01 level and I-BD being significant at the 0.05 level.

The verbal-scale IQ minus the performance-scale IQ (V-P) difference yielded significant results between the ROC and PSP groups at the 0.05 level, the ROC group exhibiting the greater V-P difference.

Comment

There are three significant topics for discussion suggested by this study: (1) cerebral dysfunction and psychopathological patterns; (2) positive spiking and clinical behavior; and (3) cerebral dysfunction and goal setting.

Cerebral Dysrhythmias and Psychopathological Patterns.—Using the NEEG group as a normative group of young psychiatric patients without detectable evidence of cerebral dysfunction and with intact family structure, separate comparisons can be made between them and the ROC and PSP groups respectively. The ROC group showed more complications of pregnancy, a trend toward speech and motor difficulties in childhood, and poorer perceptual-motor intellectual ability in young adulthood, with no significant differences in behavioral or affective functioning clinically. On the other hand, the PSP group showed significantly more childhood illness, particularly of cerebral or autonomic origin, and a trend toward neurotic or aggressive behavior or both in childhood without intellectual deficits in young adulthood. The only clinical finding was that of lowered affective response in young adulthood.

When the PSP and ROC groups were compared to each other, the ROC group showed more pregnancy complications, while the PSP group showed more neurotic or aggressive behavior or both and more significant illness in childhood. In young adulthood the ROC group had poorer perceptual-motor function while the PSP group showed poorer affective relatedness and a trend toward more aggressivity. It is clear that it is not possible to speak of psychopathological correlates of "cerebral dysrhythmia" or "borderline EEG abnormality" but that the psychiatric picture has to be clearly delineated as well as the specific EEG pattern under discussion.

The previous study of Greenberg and

Pollack⁵ showed a lower score on performance IQ, perceptual-analytic factor, and the object-assembly subtest in the ROC group when compared to the NEEG group. The current study does not fully replicate these findings although the findings are very similar. Additional analyses indicate that it is not the absolute perceptual-analytic score that is critical but rather its relative lowering when compared to verbal ability, using reliable measures such as information and object-assembly subtests of the WAIS. Moreover, the PSP patients are very similar intellectually to NEEG patients but the ROC patients have a marked difference between verbal and performance-scale IQs, unlike the PSP patients.

Positive Spiking and Clinical Behavior.

—This study reveals that, if their sociofamilial milieus are relatively intact, PSP patients have a tendency to have less affective relatedness than NEEG and ROC patients. The single variable which maintains a definite trend is that of affective relatedness which is a somewhat subtle clinical variable and not easily discerned by nonpsychiatric workers. Earlier reports of positive spiking mention, however, lack of remorse, lack of affectivity, and less anxiety in PSP patients than in those with other EEG abnormality or normal EEGs. The histories of illness identifiable as related to the central or autonomic nervous systems are in keeping with the earliest reports of Gibbs and Gibbs⁶ and others as reviewed by Henry⁸ and Hughes.⁹ Thus, the PSP patients appear to have started life with medical or neurological illness and some neurotic disability and to have become psychiatric patients with affective remoteness in early adulthood. Conversely, the ROC patients appear to have started life with a complicated pregnancy and later become psychiatric patients with perceptual-motor deficits.

Cerebral Dysfunction and Goal Setting.

—Psychiatric treatment goals for patients with cerebral dysfunction cannot be set in the same manner as for those without cerebral dysfunction. Although treatment may be very helpful, it is most likely that a patient with ROC and the clinical syndrome which usually accompanies it will not have his intellectual function in the perceptual-motor sphere improved even with the best

pharmacologic and interpersonal intervention. Counseling the patient's school, family, and prospective employer may, however, be effective in helping people in the environment to accept the patient as well as to recognize his limitations. Such intervention together with firm guidance, vocational rehabilitation, and appropriate psychotherapeutic and pharmacotherapeutic measures can often allow the patient to adapt to himself and to society.

It would appear that PSP patients can improve in their affective relatedness only to a small degree, although they and their families could benefit from the recognition of the nature of the difficulty. Thus, patients with PSP can often do well intellectually, can finish school, and can perform well at work, providing there is not excessive demand on their affective resources. Many of these patients appear to require basic instruction in the recognition of the needs of others and how their behavior may affect others, almost as if there were a cognitive deficit in the sphere of human relations.

Appropriate emphasis in treatment together with appropriate psychotropic medication may be critical in achieving clinical remission of maladaptive symptoms. It is important to work more by developing the patient's strengths and helping him and his environment to cope with his deficits then by attempting to alter the deficits.

Summary

The purposes of this study were to determine whether there are differences in incidence of developmental abnormalities among groups of hospitalized psychiatric patients between the ages of 16 and 30 with different EEG patterns and whether there are differences in behavior, affect, and cognition between patients with EEG dysrhythmia and those without EEG dysrhythmia. The patient groups consisted of 16 patients with the 14 and 6 cps positive spike pattern, 14 patients with normal EEG, and 8 patients with right posterior slowing. All patients came from relatively stable sociofamilial backgrounds and the socioeconomic status was the same for all groups.

The results showed significant differences between groups in developmental, affective, and cognitive functions. No differences be-

tween groups were found for incidences of neonatal complications, speech, and motor development, and social isolation in childhood. The ROC group had more complications of pregnancy than the other two groups and less neurotic or aggressive behavior or both in childhood. The most striking difference between groups was found in the incidence of significant illness in childhood: none of the ROC patients and only one of the NEEG patients had a serious childhood illness with overt cerebral pathological findings or autonomic endorgan pathological findings. Analyses of affective findings revealed differences in the predicted direction such that PSP patients were significantly more underemotional and had more blunted affect than did the ROC group. The ROC group showed significantly lower block-design, picture-arrangement, and object-assembly scores than did the PSP group.

These results indicate that, when family structure and socioeconomic status are controlled, there are significant differences between groups of young hospitalized psychiatric patients on the basis of cerebral dysfunction, using EEG criteria. These findings indicate that varieties of psychopathology, as well as cognitive and affective styles, may be related to cerebral function and dysfunction and that appropriate treatment programs should take these factors into account.

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Generic and Trade Names of Drug

Methypylon—*Noludar*.

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